Gramática del analizador sintáctico de Lua 5.5

Martin Moreno Jara

Notación

Los no terminales se escriben así: ⟨NON_TERMINAL⟩ y los terminales en minúsculas y sin llaves triangulares. El símbolo inicial de la gramática es es ⟨CHUNK⟩. ε es la cadena vacía, $\to$ introduce una regla y | separa sus alternativas. La gramática es LL(1).

1. Estructura del programa

Símbolo inicial, bloques y la sentencia `return`.

⟨CHUNK⟩ → ⟨BLOCK⟩

⟨BLOCK⟩ → ⟨STATLIST⟩ ⟨RETSTAT⟩

⟨STATLIST⟩ → ⟨STAT⟩ ⟨STATLIST⟩ | ε

⟨RETSTAT⟩ → `return` ⟨MAYBE_EXPLIST⟩ ⟨OPT_SEMILOCON⟩ | ε

⟨OPT_SEMILOCON⟩ → `tkn_semicolon` | ε

2. Sentencias

Las asignaciones y las llamadas se factorizan después del primer nombre.

⟨STAT⟩ → `tkn_semicolon`
| ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨LABEL⟩
| `break`
| `goto` ⟨NAME⟩
| `while` ⟨EXP⟩ ⟨DO_BLOCK_END⟩
| `repeat` ⟨BLOCK⟩ `until` ⟨EXP⟩
| ⟨IFSTAT⟩
| ⟨FORSTAT⟩
| `function` ⟨FUNCNAME⟩ ⟨FUNCBODY⟩
| ⟨LOCAL_DEC⟩
| ⟨GLOBAL_DEC⟩

⟨VAR_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩
| `tkn_period` ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨FUNCTIONCALL⟩ ⟨FUNCALL_STAT⟩
| ⟨MORE_VAR⟩ `tkn_assign` ⟨EXPLIST⟩

⟨FUNCALL_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩
| `tkn_period` ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨FUNCTIONCALL⟩ ⟨FUNCALL_STAT⟩ | ε

⟨LABEL⟩ → `tkn_goto` ⟨NAME⟩ `tkn_goto`

⟨DO_BLOCK_END⟩ → `do` ⟨BLOCK⟩ `end`

⟨IFSTAT⟩ → `if` ⟨EXP⟩ `then` ⟨BLOCK⟩ ⟨ELSEIFSTAT⟩ ⟨ELSESTAT⟩ `end`

⟨ELSEIFSTAT⟩ → `elseif` ⟨EXP⟩ `then` ⟨BLOCK⟩ ⟨ELSEIFSTAT⟩ | ε

⟨ELSESTAT⟩ → `else` ⟨BLOCK⟩ | ε

⟨FORSTAT⟩ → `for` ⟨NAME⟩ ⟨FORTYPE⟩ ⟨DO_BLOCK_END⟩

⟨FORTYPE⟩ → ⟨NUMERICAL_FOR⟩ | ⟨GENERIC_FOR⟩

⟨NUMERICAL_FOR⟩ → `tkn_assign` ⟨EXP⟩ `tkn_comma` ⟨EXP⟩ ⟨OPT_EXP⟩

⟨OPT_EXP⟩ → `tkn_comma` ⟨EXP⟩ | ε

⟨GENERIC_FOR⟩ → ⟨MORE_NAMES⟩ `in` ⟨EXPLIST⟩

3. Declaraciones

`local` y `global`.

⟨LOCAL_DEC⟩ → `local` ⟨LOCAL_OPTS⟩

⟨LOCAL_OPTS⟩ → ⟨FUNC_DEC⟩ | ⟨ATTNAME_BRANCH⟩

⟨ATTNAME_BRANCH⟩ → ⟨ATTNAMELIST⟩ ⟨OPT_EXPLIST⟩

⟨GLOBAL_DEC⟩ → `global` ⟨GLOBAL_OPTS⟩

⟨GLOBAL_OPTS⟩ → ⟨FUNC_DEC⟩ | ⟨ATTNAMELIST⟩ ⟨OPT_EXPLIST⟩

⟨FUNC_DEC⟩ → `function` ⟨NAME⟩ ⟨FUNCBODY⟩

⟨OPT_EXPLIST⟩ → `tkn_assign` ⟨EXPLIST⟩ | ε

⟨ATTNAMELIST⟩ → ⟨NAME⟩ ⟨ATTRIB⟩ ⟨NAME_ATTRIB_LIST⟩

⟨NAME_ATTRIB_LIST⟩ → `tkn_comma` ⟨NAME⟩ ⟨ATTRIB⟩ ⟨NAME_ATTRIB_LIST⟩ | ε

⟨ATTRIB⟩ → `tkn_less` ⟨NAME⟩ `tkn_greater` | ε

4. Funciones

Nombres de función, cuerpos y listas de parámetros.

⟨FUNCNAME⟩ → ⟨NAME⟩ ⟨DOTTED_FUNCNAME⟩ ⟨COLON_FUNCNAME⟩

⟨DOTTED_FUNCNAME⟩ → `tkn_period` ⟨NAME⟩ ⟨DOTTED_FUNCNAME⟩ | ε

⟨COLON_FUNCNAME⟩ → `tkn_colon` ⟨NAME⟩ | ε

⟨FUNCTIONDEF⟩ → `function` ⟨FUNCBODY⟩

⟨FUNCBODY⟩ → `tkn_opening_par` ⟨OPT_PARLIST⟩ `tkn_closing_par` ⟨BLOCK⟩ `end`

⟨OPT_PARLIST⟩ → ⟨PARLIST⟩ | ε

⟨PARLIST⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩ | ⟨VARPARAGRAM⟩

⟨PARLIST_TAIL⟩ → `tkn_comma` ⟨PARLIST_AFTER_COMMA⟩ | ε

⟨PARLIST_AFTER_COMMA⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩ | ⟨VARPARAGRAM⟩

⟨VARPARAGRAM⟩ → `tkn_varargs` ⟨OPT_NAME⟩

⟨OPT_NAME⟩ → ⟨NAME⟩ | ε

5. Listas

Listas de variables, de nombres y de expresiones.

⟨MORE_NAMES⟩ → `tkn_comma` ⟨NAME⟩ ⟨MORE_NAMES⟩ | ε

⟨MORE_VAR⟩ → `tkn_comma` ⟨PREFIXEXP⟩ ⟨MORE_VAR⟩ | ε

⟨MAYBE_EXPLIST⟩ → ⟨EXPLIST⟩ | ε

⟨EXPLIST⟩ → ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩

⟨MORE_EXP⟩ → `tkn_comma` ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩ | ε

⟨EXPLIST_ITEM⟩ → ⟨EXP⟩ | ⟨FUNCTIONDEF⟩

6. Expresiones

Operandos, expresiones prefijo y llamadas.

⟨EXP⟩ → ⟨OPERAND⟩ ⟨EXP_PRIME⟩

⟨EXP_PRIME⟩ → ⟨ARITH_TAIL⟩ ⟨CMP_TAIL⟩ ⟨LOGIC_TAIL⟩

⟨OPERAND⟩ → ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩ | ⟨OPERAND_NOT_ID⟩

⟨OPERAND_NOT_ID⟩ → `nil` | `false` | `true`
| `tkn_num` | `tkn_str` | ⟨TABLECONSTRUCTOR⟩
| ⟨UNOP⟩ ⟨OPERAND⟩ | `tkn_opening_par` ⟨EXP⟩ `tkn_closing_par`

⟨PREFIXEXP⟩ → ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩

⟨PREFIXEXP_PRIME⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨PREFIXEXP_PRIME⟩
| `tkn_period` ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩
| ⟨FUNCTIONCALL⟩ ⟨PREFIXEXP_PRIME⟩ | ε

⟨FUNCTIONCALL⟩ → ⟨ARGS⟩ | `tkn_colon` ⟨NAME⟩ ⟨ARGS⟩

⟨ARGS⟩ → `tkn_opening_par` ⟨MAYBE_EXPLIST⟩ `tkn_closing_par`

7. Tablas

Constructores de tablas.

⟨TABLECONSTRUCTOR⟩ → `tkn_opening_key` ⟨FIELDLIST⟩ `tkn_closing_key`

⟨FIELDLIST⟩ → ⟨FIELD⟩ ⟨FIELD_TAIL⟩ | ε

⟨FIELD_TAIL⟩ → ⟨FIELDSEP⟩ ⟨FIELDLIST⟩ | ε

⟨FIELD⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra tkn_assign` ⟨FIELD_VALUE⟩
| ⟨NAME⟩ ⟨FIELD_ID⟩
| ⟨OPERAND_NOT_ID⟩ ⟨EXP_PRIME⟩
| ⟨FUNCTIONDEF⟩

⟨FIELD_ID⟩ → `tkn_assign` ⟨FIELD_VALUE⟩
| ⟨PREFIXEXP_PRIME⟩ ⟨EXP_PRIME⟩

⟨FIELD_VALUE⟩ → ⟨EXP⟩ | ⟨FUNCTIONDEF⟩

⟨FIELDSEP⟩ → `tkn_comma` | `tkn_semicolon`

8. Operadores

Operadores binarios por nivel, operadores unarios y nombres.

$$
\begin{aligned}
\langle \text{LOGIC_TAIL} \rangle &\rightarrow \langle \text{LOGICAL_OP} \rangle \ \langle \text{CMP_EXP} \rangle \ \langle \text{LOGIC_TAIL} \rangle \ \mid \ \varepsilon \\
\langle \text{CMP_EXP} \rangle &\rightarrow \langle \text{ARTH_EXP} \rangle \ \langle \text{CMP_TAIL} \rangle \\
\langle \text{CMP_TAIL} \rangle &\rightarrow \langle \text{RELATIONAL_OP} \rangle \ \langle \text{ARTH_EXP} \rangle \ \mid \ \varepsilon \\
\langle \text{ARTH_EXP} \rangle &\rightarrow \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \\
\langle \text{ARITH_TAIL} \rangle &\rightarrow \langle \text{FLAT_OP} \rangle \ \langle \text{OPERAND} \rangle \ \langle \text{ARITH_TAIL} \rangle \ \mid \ \varepsilon \\
\langle \text{FLAT_OP} \rangle &\rightarrow \langle \text{ARITH_OP} \rangle \mid \langle \text{BITWISE_OP} \rangle \mid \langle \text{OTHER_OP} \rangle \\
\langle \text{ARITH_OP} \rangle &\rightarrow \texttt{tkn_plus} \mid \texttt{tkn_minus} \mid \texttt{tkn_times} \mid \texttt{tkn_div} \\
&\quad \mid \texttt{tkn_floor_div} \mid \texttt{tkn_power} \mid \texttt{tkn_mod} \\
\langle \text{RELATIONAL_OP} \rangle &\rightarrow \texttt{tkn_less} \mid \texttt{tkn_leq} \mid \texttt{tkn_greater} \\
&\quad \mid \texttt{tkn_geq} \mid \texttt{tkn_equal} \mid \texttt{tkn_neq} \\
\langle \text{LOGICAL_OP} \rangle &\rightarrow \texttt{and} \mid \texttt{or} \\
\langle \text{BITWISE_OP} \rangle &\rightarrow \texttt{tkn_bit_and} \mid \texttt{tkn_bitex_or} \mid \texttt{tkn_bit_or} \\
&\quad \mid \texttt{tkn_right_shift} \mid \texttt{tkn_left_shift} \\
\langle \text{OTHER_OP} \rangle &\rightarrow \texttt{tkn_concat} \\
\langle \text{UNOP} \rangle &\rightarrow \texttt{tkn_minus} \mid \texttt{not} \mid \texttt{tkn_length} \\
\langle \text{NAME} \rangle &\rightarrow \texttt{id} \mid \texttt{print} \mid \texttt{pcall} \\
&\quad \mid \texttt{error} \mid \texttt{warn} \mid \texttt{xpcall}
\end{aligned}
$$